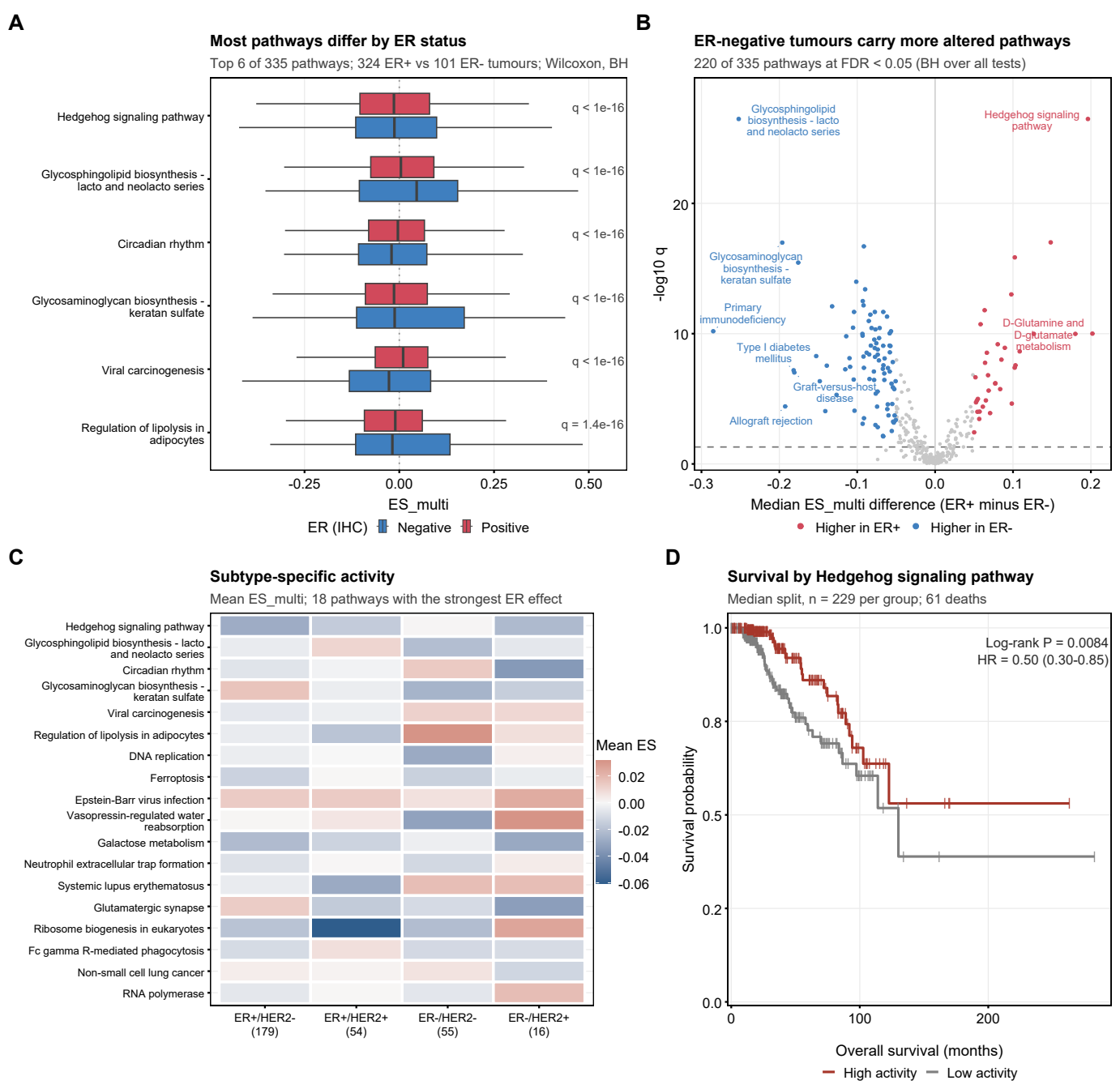

A
Most pathways differ by ER status
Top 6 of 335 pathways; 324 ER+ vs 101 ER- tumours; Wilcoxon, BH
Hedgehog signaling pathway
Glycosphingolipid biosynthesis - lacto and neolacto series
Circadian rhythm
Glycosaminoglycan biosynthesis - keratan sulfate
Viral carcinogenesis
Regulation of lipolysis in adipocytes
q < 1e-16
q < 1e-16
q < 1e-16
q < 1e-16
q < 1e-16
q = 1.4e-16
-0.25
0.00
0.25
0.50
ES_multi
ER (IHC)
Negative
Positive
B
ER-negative tumours carry more altered pathways
220 of 335 pathways at FDR < 0.05 (BH over all tests)
Glycosphingolipid biosynthesis - lacto and neolacto series
Hedgehog signaling pathway
Glycosaminoglycan biosynthesis - keratan sulfate
Primary immunodeficiency
Type I diabetes mellitus
Graft-versus-host disease
Allograft rejection
D-Glutamine and D-glutamate metabolism
-log10 q
20
10
0
-0.3
-0.2
-0.1
0.0
0.1
0.2
Median ES_multi difference (ER+ minus ER-)
Higher in ER+
Higher in ER-
C
Subtype-specific activity
Mean ES_multi; 18 pathways with the strongest ER effect
Hedgehog signaling pathway
Glycosphingolipid biosynthesis - lacto and neolacto series
Circadian rhythm
Glycosaminoglycan biosynthesis - keratan sulfate
Viral carcinogenesis
Regulation of lipolysis in adipocytes
DNA replication
Ferroptosis
Epstein-Barr virus infection
Vasopressin-regulated water reabsorption
Galactose metabolism
Neutrophil extracellular trap formation
Systemic lupus erythematosus
Glutamatergic synapse
Ribosome biogenesis in eukaryotes
Fc gamma R-mediated phagocytosis
Non-small cell lung cancer
RNA polymerase
ER+/HER2- (179)
ER+/HER2+ (54)
ER-/HER2- (55)
ER-/HER2+ (16)
Mean ES
0.02
0.00
-0.02
-0.04
-0.06
D
Survival by Hedgehog signaling pathway
Median split, n = 229 per group; 61 deaths
Log-rank P = 0.0084
HR = 0.50 (0.30-0.85)
Survival probability
1.0
0.8
0.5
0.2
0.0
0
100
200
Overall survival (months)
High activity
Low activity